Demo: Meridian Works

Network summary report

Prepared on 22 October 2026

40 members, 4 relationship layers analysed.

Ties are read in the direction they were rated (directed view).

Composite weights: set by the analyst.

Names are hidden: members appear as team, level and number codes, and names in free text are replaced.

This report describes the structure of working relationships. It is not an assessment of anyone, and should not be used to evaluate individual performance.

About the data

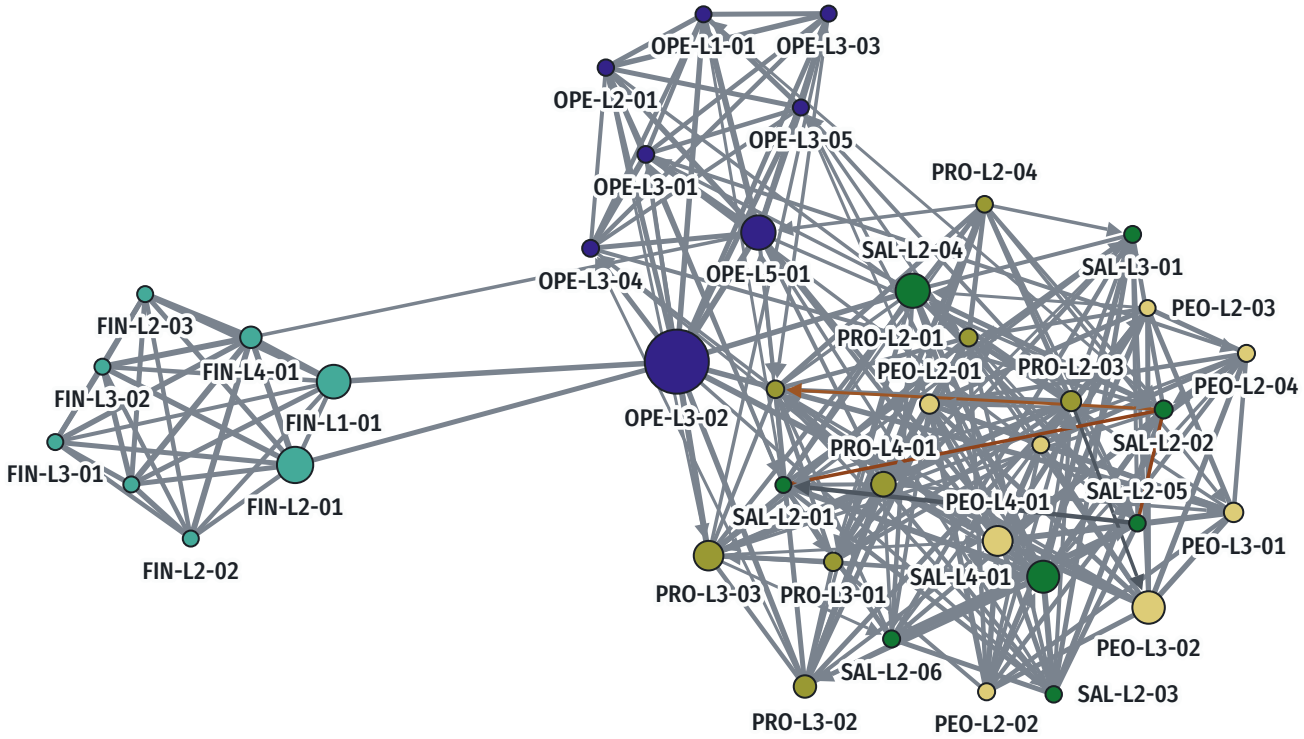
Coverage 95%, at or above the 80% threshold. Two of 40 people gave fewer than half of the ratings possible. What these two people's own ratings would show is missing; their results rest largely on other people's reports. Of the 708 ties reported where the other person also answered, 18% were not reported back.

Contents

Network map 1
Key network metrics 2
Insights 4
Method notes 7
Data coverage 10
Ethics statement 11

Network map

Ties drawn from Composite of all layers. The legend explains every encoding.



Node size
Bridging
0.000 0.222 0.443

Node fill
Team or function
- Finance
- Operations
- Product
- Sales
- People

Edge width
Composite of all layers
0.20 0.60 1.00
Ties below 0.30 hidden
Every metric uses all ties, hidden or not

Edge colour
Valence: negative ratings only
-3 -1
Neutral or positive
Valence not rated

Direction
One way, from rater to rated member
Both ways: one line, the two directions averaged

Drawn for FIN-L4-01.

Key network metrics

Whole-network measures for the composite and each layer. Definitions and caveats are in the method notes.

Measure	Composite of all layers	Connection strength	Valence, positive ratings	Valence, negative ratings	Informal collaboration
Members	40	40	40	40	40
Ties	715	546	597	11	543
Density	0.458	0.350	0.383	0.007	0.348
Ties returned (tie reciprocity)	0.817	0.696	0.596	0.909	0.729
Average clustering	0.344	0.285	0.233	0.077	0.299
Separate parts (weakly connected components)	1	1	1	–	1
Members in the largest part (weak)	40	40	40	–	40
Centralisation of connections in	0.187	0.141	0.133	0.072	0.143
Centralisation of connections out	0.266	0.167	0.212	0.072	0.195
Centralisation of bridging	0.170	0.327	0.012	–	0.323
Communities (Louvain)	4	5	5	–	4
Modularity	0.457	0.460	0.378	–	0.414

Measure	Formal collaboration
Members	40
Ties	535
Density	0.343
Ties returned (tie reciprocity)	0.789
Average clustering	0.410
Separate parts (weakly connected components)	1
Members in the largest part (weak)	40
Centralisation of connections in	0.227
Centralisation of connections out	0.201
Centralisation of bridging	0.202
Communities (Louvain)	4
Modularity	0.497

For negative ties, "–" marks a measure that is not reported. Separate parts, communities, modularity and the centralisation of bridging describe how a network holds together, which negative ties do not: with few of them, nearly every member is a part and a community of their own.

Composite as calculated C = (0.33·r(Connection strength) + 0.33·r(Informal collaboration) + 0.33·r(Formal collaboration)) × (1 + 0.5·s(Valence)); C <= 1

Ties within and between groups by team or function, on Composite of all layers

Group	Members	Inside	Leaving	E-I index
Finance	8	56	6	−0.81
Operations	8	55	184	0.54
Product	8	56	256	0.64
Sales	8	56	244	0.63
People	8	56	182	0.53
All groups	40	279	436	0.22

E-I runs from −1 (every tie stays inside the group) to +1 (every tie leaves it). Larger groups have more chances of internal ties.

Structural balance: Valence

Measure	Value
Balanced trios	1,573
Unbalanced trios	79
Balance ratio	0.952

Formal and informal collaboration, pairs

Measure	Value
Pairs with formal ties only	101
Pairs with informal ties only	108
Pairs with formal and informal ties	419
Pairs with neither	819
Pairs not classified (a rating is missing)	113

Insights

Rule-based observations from this analysis. Each is a question to explore with the people involved, not a finding about anyone.

Potential brokers

Rule Bridging (betweenness) on the composite among the highest 10% of members and at least 0.05, and constraint at or below the median. At most 5 members, highest bridging first.

Three members lie on many of the shortest routes between colleagues, while their own contacts are not closely tied to one another. What would happen to those routes if one of them were unavailable?

Members OPE-L3-02, OPE-L5-01, SAL-L2-04

Two of the three share team or function Operations. Two of the three share tenure band Over 5 years.

Shown with Ties from composite of all layers, node size by bridging, force-directed layout, three members highlighted.

Peripheral members

Rule Received strength at or below the 20th percentile of members and at least 10% below the median, on more than half of the layers considered: each unsigned layer except the conflict layers, and the positive ratings of each signed layer, not the composite. A layer on which no one receives a tie is not considered. At most 5 members are listed, furthest below the median first. Members who make up at least 50% of a silo group are named with that group instead.

Six members this rule found are part of a group listed under Silos, and named there.

Possible overload

Rule On the advice and workflow dependency layers: connections in at or above the 90th percentile of members and at least 2 times the median. At most 5 members per layer.

Not applied: this rule reads the advice and workflow dependency layers. Enable either layer and add its ratings.

Silos

Rule Members are grouped by team or function. A group of at least 3 members whose E-I index is −0.50 or lower on the composite or on any unsigned layer except the conflict layers, which means at least three ties inside the group for every tie outside it, counting only layers with at least 5 ties involving the group. The layer with the lowest index is quoted. Not applied when fewer than 2 groups have 3 members or more.

On informal collaboration, most ties involving Finance stay within Finance: 54 inside and 3 outside (E-I index −0.89). Six of its 8 members also receive fewer ties than most colleagues on most layers (see Peripheral members). How does work reach the rest of the organisation?

Members FIN-L4-01, FIN-L3-01, FIN-L2-01, FIN-L2-02, FIN-L3-02, FIN-L1-01, FIN-L3-03, FIN-L2-03

Shown with Ties from informal collaboration, node size by bridging, grouped by team or function, eight members highlighted.

Negative clusters

Rule Pairs who rate each other negatively in both directions on a signed layer, joined into groups through shared members. A group of at least 3 members is shown. Read from the directed ratings in either view.

In a group of 4 members, 5 pairs rate each other negatively on valence, negative ratings in both directions. What shared circumstances might lie behind these ratings?

Members PRO-L2-02, SAL-L2-01, SAL-L2-02, SAL-L2-05

All four share level L2. Three of the four share team or function Sales.

Shown with Ties from composite of all layers, node size by bridging, force-directed layout, four members highlighted.

Negative ratings not returned

Rule On each signed layer, negative ratings whose reverse rating is not negative: neutral, positive or not given. A layer with at least 3 is shown, with counts by team or function, the rater's group first. Read from the directed ratings in either view.

No observation: nothing in this analysis meets the rule.

Formal ties with no informal counterpart

Rule Members are grouped by team or function. For every pair of groups, and every group on its own: at least 50% of formal collaboration ties have no informal collaboration tie, and there are at least 5 such ties. Pairs with a missing rating on either layer are left out. The 5 members in the most such ties are highlighted.

Between Product and Sales, 85 of 119 formal collaboration ties (71%) have no informal counterpart. What happens between these people outside formal processes?

Members PRO-L2-01, SAL-L2-02, PRO-L3-02, PRO-L2-04, SAL-L4-01

Three of the five share level L2. Three of the five share tenure band 1 to 3 years.

Shown with Ties from formal collaboration, node size by connections in, grouped by team or function, showing team or function Product or Sales, five members highlighted.

Informal ties with no formal counterpart

Rule Members are grouped by team or function. For every pair of groups, and every group on its own: at least 50% of informal collaboration ties have no formal collaboration tie, and there are at least 5 such ties. Pairs with a missing rating on either layer are left out. The 5 members in the most such ties are highlighted.

Between People and Product, 35 of 57 informal collaboration ties (61%) have no formal counterpart. What work do these ties carry that the formal structure does not show?

Members PEO-L2-02, PRO-L2-03, PRO-L3-01, PRO-L2-01, PEO-L2-01

Four of the five share level L2. Three of the five share location Glasgow.

Shown with Ties from informal collaboration, node size by connections in, grouped by team or function, showing team or function People or Product, five members highlighted.

Caveats The thresholds are conventions chosen for this tool, not published standards; they are stated wherever a rule is shown so a reader can judge them. Every rule inherits the caveats of the metric it reads (sections 3 to 6): betweenness assumes shortest-path flow, the E-I index counts ties and favours larger groups, received strength depends on who answered. A negative rating not returned negatively may reflect a real difference in experience, a rater who rates everyone low or high, or a colleague who did not answer; the signed rule counts ratings, not people. The rules describe positions in the network,

which have many causes (role, tenure, location, workload, the survey itself); an observation is a reason to ask, not an answer. Below the coverage threshold the panel repeats the coverage warning.

Method notes

Drawn from Graticule's method notes, which give every formula in full with its references. Only the measures used in this report are included.

Not rated is not zero

A rating of 0 means "no connection" or "never" (on a signed layer, "neutral"). A missing rating means the person did not answer. The two are never treated alike: a missing rating does not create a tie, is not counted as a 0 in averages, and is counted as missing in data coverage. Inside the engine a missing rating is the value NaN; in project files it is null or an absent tie.

A third state, does not apply, records that the rater said the question has no answer for that colleague, such as the quality of a relationship they do not have. It is neither 0 nor missing. Metrics treat it like a missing rating (no tie), but coverage leaves it out of the ratings that were possible, because there was nothing to give. In project files it is null with not_applicable: true; in ties files and matrix entry it is written n/a.

Rescaling

Every layer is rescaled to 0–1 before any metric, so values from different layers can be compared and combined.

Unsigned layers: r = (v − min) / (max − min). On a 0–5 scale, 0 to 0, 1 to 0.2, 5 to 1.

Signed layers (valence, energy): s = v / max for v >= 0 and v / |min| for v < 0, so s runs from −1 to +1. Each signed layer is split into a positive sub-layer, w+ = max(s, 0), and a negative sub-layer, w− = max(−s, 0). The two are analysed separately and never netted.

Caveat: rescaling assumes the points of a rating scale are equally spaced. A "5" is treated as five times a "1".

Directed and symmetrised views

In the directed view A's rating of B and B's rating of A are separate ties.

In the symmetrised view each pair has one tie, combined from the two directions by the rule the analyst chooses:

mean: (w_ij + w_ji) / 2;

minimum: min(w_ij, w_ji), a tie only if both sides report one;

maximum: max(w_ij, w_ji), a tie if either side reports one.

If only one of the two has rated the other, that rating is used on its own (mean, minimum and maximum then agree). If neither has, the pair is missing. Signed layers are combined before they are split, so when A rates B positively and B rates A negatively the pair's sign follows the rule: the sign of the mean, negative under minimum, positive under maximum.

Caveat: the minimum rule makes the network sparser and the maximum rule denser; metrics can change markedly between them. The chosen rule is part of every result.

Composite

Meaning One combined relationship score per pair, weighting the layers by importance.

Caveats The composite is only as meaningful as the weights: it is a statement of priorities, not a measurement. Re-normalising over rated layers means a pair rated on one layer only is scored on that layer alone.

Density

Meaning The share of possible ties that exist.

Caveats Ignores strength. Not comparable across networks of very different size: large networks are sparser because people have limited time.

Reciprocity

Meaning How often ties are returned: when one member reports a tie to another, how often the other reports one back.

Caveats The two figures differ: a network with one mutual pair and one one-way pair has tie reciprocity 2/3 and dyad reciprocity 1/2. Both ignore tie strength. Missing ratings lower reciprocity, because an unanswered survey looks like an unreturned tie.

Average clustering

Meaning How clustered the network is overall: the mean of the members' local clustering.

Caveats Members with fewer than two contacts pull the average down.

Components

Meaning Groups of members connected by some chain of ties, with no ties to other groups.

Caveats An isolated member is a group of one.

Communities (Louvain) and modularity

Meaning Groups of members more densely tied to each other than to the rest, found automatically.

Caveats Louvain is a heuristic: it finds a good partition, not necessarily the best, and a different node order or seed can give a different one of similar quality. Graticule uses graphology's implementation, whose partitions can differ from NetworkX's; on every reference case its modularity is equal to or higher than NetworkX's. With a single run, or with graphology's default fast local moving, it fell up to 0.015 short. Modularity around 0.3 or above is usually read as clear community structure; values near 0 mean none. The resolution gamma = 1 is the same default as NetworkX; larger values find smaller communities.

Centralisation

Meaning How much the network is dominated by one member, compared with a star, the most centralised shape possible.

Caveats Computed for binary degree and betweenness only, where the star-graph maximum is established. Not defined for fewer than three members. NetworkX has no centralisation function; the reference values are computed from NetworkX degree and betweenness.

E-I index

Meaning Whether ties stay within groups (such as teams) or cross between them.

Caveats Counts ties, not strength. Larger groups have more internal opportunities, so their index is lower even when members mix freely; compare with cross-group density. Members with no value for the attribute are left out, and the number left out is reported.

Structural balance

Meaning Whether trios of members are in a stable pattern. Balance theory (Heider, 1946; Cartwright & Harary, 1956) holds that "my friend's friend is my friend" (+++) and "my enemy's enemy is my friend" (+−−) are stable, while a trio with one negative pair (++−) or three (−−−) is under strain.

Caveats Trios with a neutral (0) or unrated pair are not counted. The dyad sign depends on the symmetrisation rule when the two directions disagree.

Formal–informal comparison

Meaning Classifies every pair as formal only, informal only, both, or neither, using the formal and informal collaboration layers.

Caveats "Formal only" means the collaboration is required but not reported as happening outside formal channels; it is a prompt for inquiry, not evidence of a problem.

Data coverage

Meaning The share of possible ratings that were given, per rater and overall.

Data coverage

Coverage is 95%, at or above the 80% threshold.

5,938 of 6,240 possible ratings were given.

Ratings	Value
Given	5,938
Declined	10
Not entered	292
Does not apply	0
Possible	6,240
Overall response rate	95%

The share of possible ratings that were given, across enabled layers. A rating of 0 counts as given; declined and never-entered ratings do not. A rating marked “does not apply” was never possible, so it is left out altogether.

2 raters are below 80%:

Rater	Response rate
SAL-L3-01	47%
OPE-L3-04	49%

Where the ratings came from

Source	Value
Not recorded	5,948

Not recorded: the project does not say how these ratings were collected. Graticule records whether a rating came from a survey response, an imported file or entry in the matrix; ratings written by older versions or by other tools, and the demo’s ratings, carry no source. They are analysed like every other rating.

Survey response rates

No survey was run in Graticule for this project; ratings were imported or entered.

Ethics statement

This report maps working relationships between named people. It describes the structure of a network; it does not measure anyone's performance, and it should not be used in performance evaluation or in decisions about individuals.

Participants should have given informed consent, and should have been told what would be collected, who would see the results and how the results would be used. Survey responses are confidential, not anonymous: the analyst sees who gave which ratings.

Processing named employee data is likely to engage data protection law. Store, share and delete this report as personal data, following the advice your organisation has taken.

Every position in a network has many causes, including role, tenure, location, workload and the survey itself. Observations here are prompts for conversation with the people involved, not conclusions about them.

Names in this report are replaced by team, level and number codes. Codes of small teams can still identify someone, so treat the report as personal data.